

Konsep, Strategi, Metode dan Target Eksplorasi Geofisika

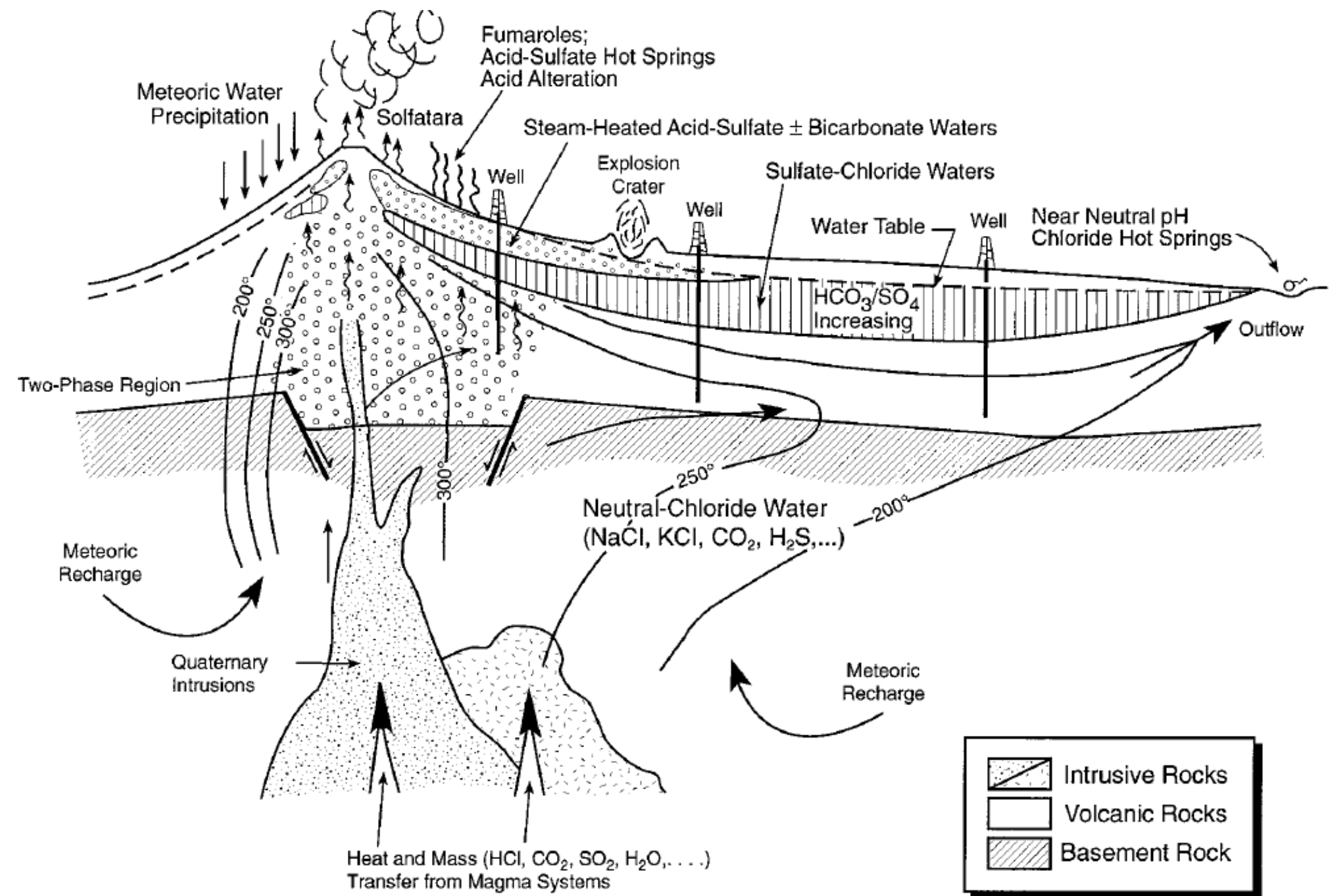
Fauziah Maswah S.Si., M.T.

25 November – 6 Desember 2024
Zoom dan PLTA Bangkok

Sistem Geotermal

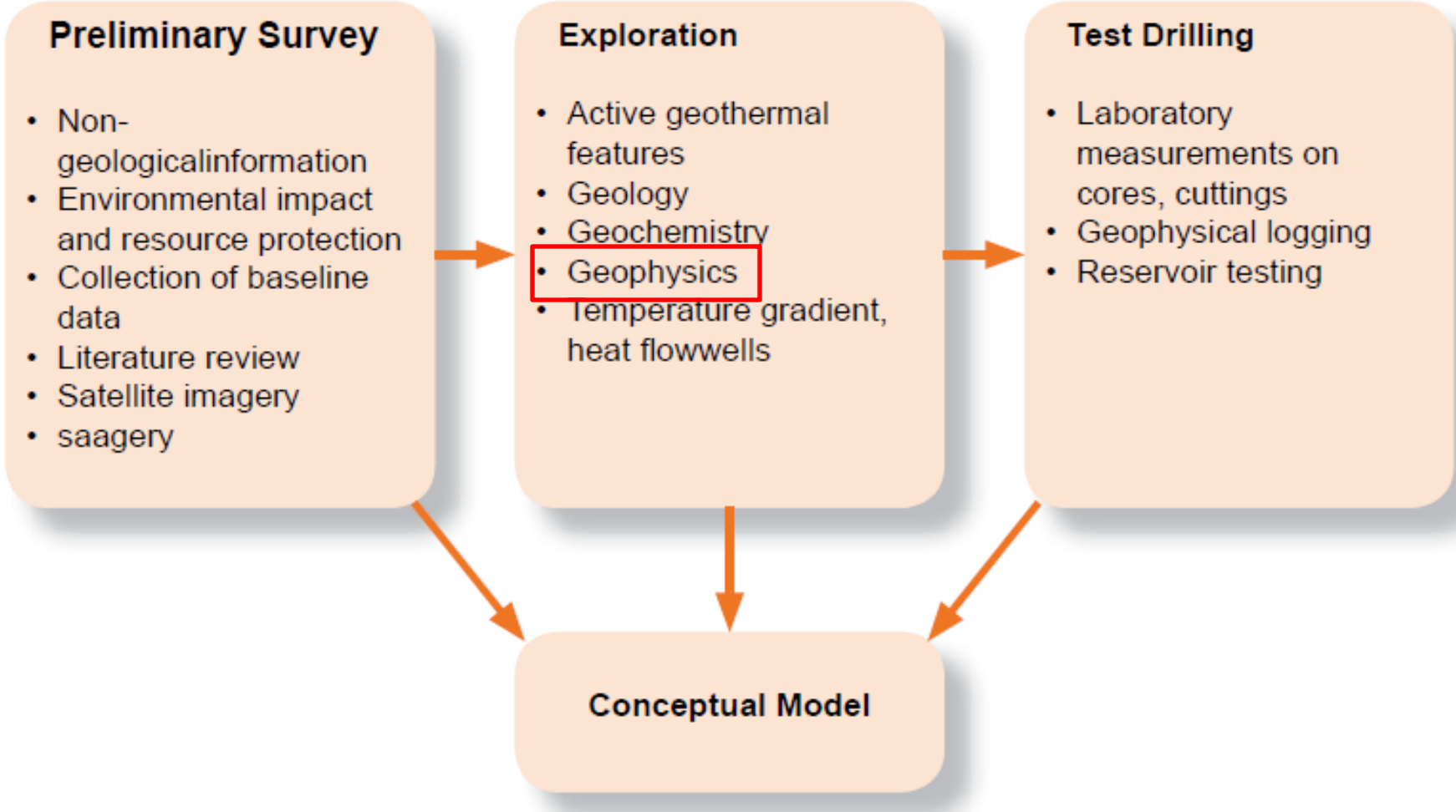
Komponen sistem geotermal
(Hochstein dan Browne, 2000):

- Sumber panas
- Reservoir
- *Recharge* dan *Discharge*



Goff dan Janik, 2000)

Tahap Eksplorasi Geofisika



GEOLOGICAL AND SURFACE STUDIES	GEOCHEMICAL SURVEYS	GEOFYSICAL SURVEYS
• Mapping surface geology (Puente and De La Peña, 1979) • Locating and mapping active geothermal surface features • Structural geological interpretation • Earthquake locations and focal mechanisms	• Collect samples from all thermal features for analysis • Geothermometry (water and gases) (Ellis, 1979; Giggenbach and Goguel, 1989) • pH + electrical conductivity • Flow rate and temperature of fluids discharging from active thermal features • Soil sampling and gas flux (Harvey et al., 2011)	• Remote sensing • Heat flow survey of fluid discharge sites (Fisher, 1964) • Gravity and magnetics (Pálmason, 1975) • Electrical resistivity • Magnetotellurics (Anderson et al., 2000) • Passive seismic monitoring • 2D and 3D seismic reflection • Temperature gradient and conductive heat flow

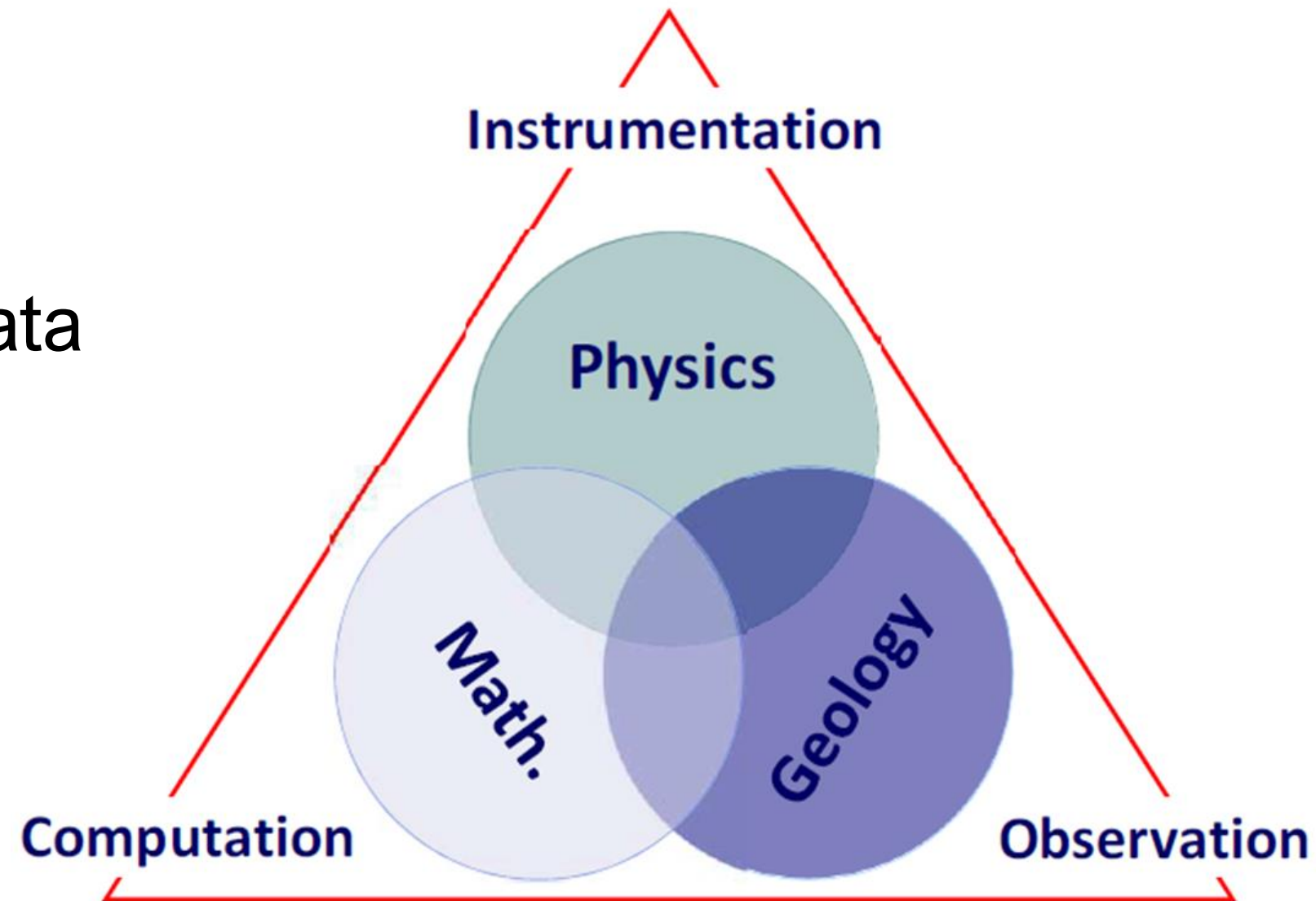
Biaya Eksplorasi

c. Penyelidikan dan Penelitian Geofisika			
1) Gravity			
a) Regional > 1 km, Minimal 30 Titik	per titik	Rp	1.500.000,00
b) Detil < 100 m, Minimal 100 Titik	per titik	Rp	250.000,00
2) Magnetik			
a) Regional > 1 km, Minimal 50 Titik	per titik	Rp	1.000.000,00
b) Detil < 100 m, Minimal 100 Titik	per titik	Rp	100.000,00
3) Seismik Refraksi (Sumber Daya Palu)	per km lintasan	Rp	20.000.000,00
4) Magnetotelurik (Mt) (Minimal 10 Titik)	per titik	Rp	25.000.000,00
5) Geolistrik Multi Channel (Minimal 2 Lintasan)	per km lintasan	Rp	20.000.000,00

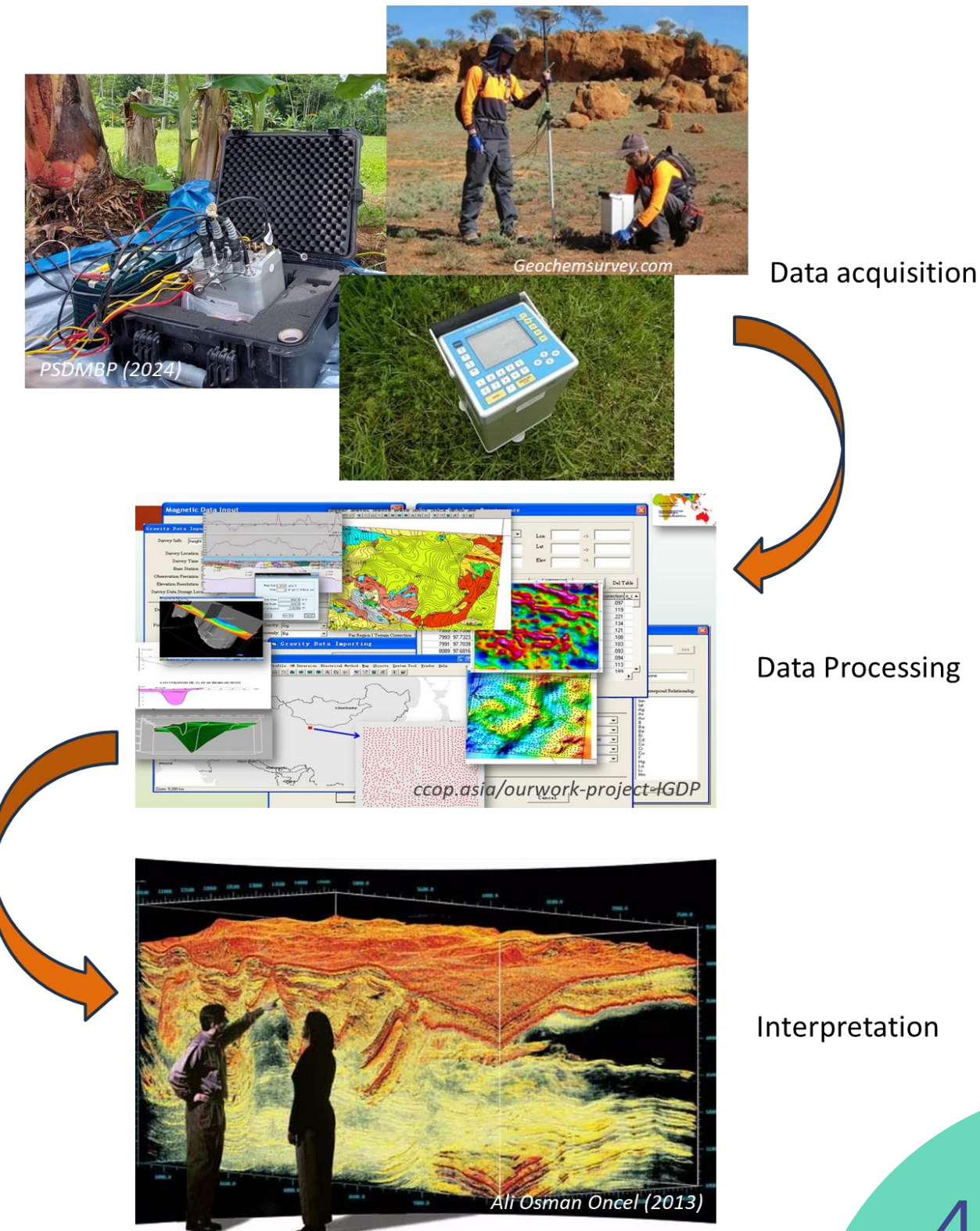
(ESDM, 2019)

Ruang Lingkup Geofisika

- Akuisisi Data
- Pengolahan Data
- Pemodelan
- Interpretasi

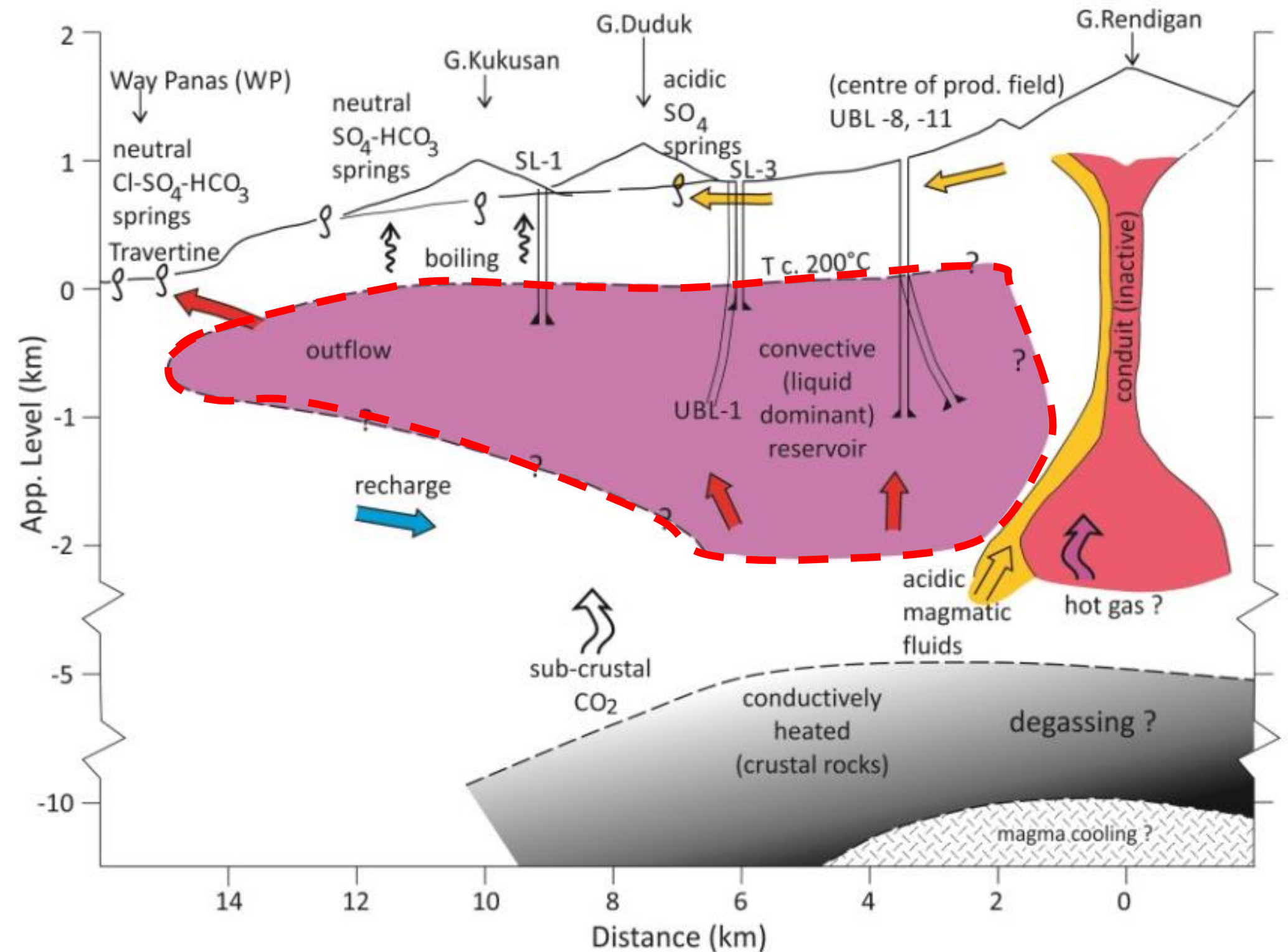


Source: Hendra Grandis



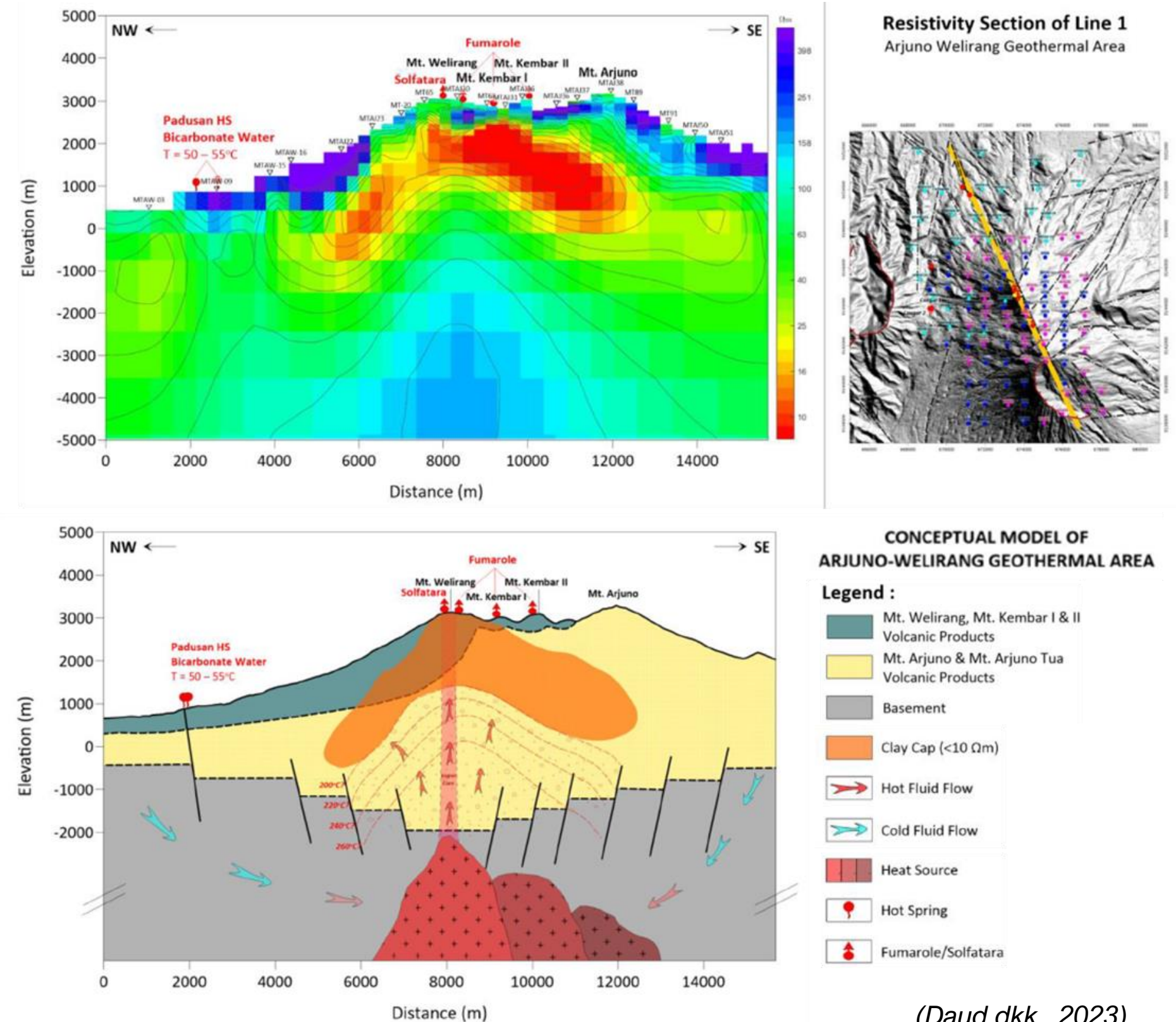
Target

- Geometri
- Litologi
- Model Konseptual
- Estimasi Potensi



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(Daud dkk., 2023)

Metode:

- Controlled-Source Electromagnetics (CSEM)
 - Electrical resistivity (DC)
 - Gravity
 - Magnetic
 - Magnetotelluric (MT)
 - Microearthquake
- Metode Aktif
- Metode Passif

Reconnaissance

- Gravity (dan Magnetik)
- Geolistrik (Sounding dan Mapping)

Advance

- Magnetotelluric (MT)
- TDEM (Time Domain Electromagnetic)

Pengembangan (*Development*)

- Microgravity
- Microearthquake

Metode Geofisika dan Properti Fisika yang Terukur

Metode Geofisika	Properti Fisika
Gaya Berat (Gravity)	Densitas (Density)
Geomagnet (Magnetic)	Suseptibilitas Magnetik (Magnetic Suseptibility)
Magnetotellurik (MT)	Electrical Resistivity
Geolistrik (Geo-electrical)	Electrical Resistivity
Seismik (Seismic)	Kecepatan Geolombang Seismik (Seismic Wave Velocity)

TERIMA KASIH



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